

Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	9 March 2026
Team ID	SWUID202641986
Project Name	Project - Automobile Insurance Fraud Detection System
Maximum Marks	8 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Registration	USN-1	As a user, I can register into the system by entering my details.	2	High	Viraj Potdar
Sprint-1	Login	USN-2	As a user, I can login into the system using email and password.	1	High	Viraj Potdar
Sprint-1	Dataset Upload	USN-3	As an admin, I can upload insurance claim dataset for analysis.	2	Medium	Samruddhi Yadav
Sprint-2	Data Preprocessing	USN-4	As a developer, I can clean and preprocess the dataset before training the model.	2	High	Snehal Khot
Sprint-2	Feature Selection	USN-5	As a developer, I can select important features for fraud detection.	1	High	Samruddhi Yadav
Sprint-3	Model Training	USN-6	As a system, I can train machine learning models using claim data.	3	High	Tanuja Talekar
Sprint-3	Model Comparsion	USN-7	As a developer, I can compare different ML algorithms to find the best model.	2	Medium	Tanuja & Snehal

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Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	24 Oct 2022	29 Oct 2022	20	29 Oct 2022
Sprint-2	20	6 Days	31 Oct 2022	05 Nov 2022	20	05 Nov 2022
Sprint-3	20	6 Days	07 Nov 2022	12 Nov 2022	20	12 Nov 2022
Sprint-4	20	6 Days	14 Nov 2022	19 Nov 2022	20	19 Nov 2022

Velocity:

Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

$$AV = \frac{\textit{sprint duration}}{\textit{velocity}} = \frac{20}{10} = 2$$

Velocity = Total Story Points / Sprint Duration

Velocity = 13/10

Velocity~ 1.3 points/day

Burndown Chart:

A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over time.

Day	Ideal Remaining Points	Actual Remaining Points (Projected)	Task Completed
0	13	13	Project Start
2	11.3	10	Registration & Login (Viraj)
4	9.5	8	Dataset Upload (Samruddhi)
7	6.9	5	Preprocessing & Features (Snehal)
11	3.5	2	Model Training (Tanuja)
15	0	0	Model Comparison & Finalization

<https://www.visual-paradigm.com/scrum/scrum-burndown-chart/>

<https://www.atlassian.com/agile/tutorials/burndown-charts>

Reference:

Agile Burndown Chart Guide: <https://www.atlassian.com/agile/tutorials/burndown-charts>

Scrum Framework & Sprints: <https://scrumguides.org/scrum-guide.html>

Mural Design Thinking Templates: <https://www.mural.co/templates/empathy-map-canvas>

Scikit-Learn Random Forest Documentation: <https://scikit-learn.org/stable/modules/ensemble.html>